

# RCP Simulation Visualization

## Setup

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)

packages <- c("ggplot2", "dplyr", "patchwork")
invisible(lapply(packages, library, character.only = TRUE))

find_project_root <- function(start = getwd()) {
  path <- normalizePath(start, mustWork = FALSE)
  for (i in seq_len(8)) {
    if (dir.exists(file.path(path, "Code")) && dir.exists(file.path(path, "Data"))) return(path)
    parent <- dirname(path)
    if (identical(parent, path)) break
    path <- parent
  }
  stop("Set LC_DLNM_BASE_DIR to the project root.")
}

base_dir <- Sys.getenv("LC_DLNM_BASE_DIR")
if (!nzchar(base_dir)) base_dir <- find_project_root()
invisible(lapply(list.files(file.path(base_dir, "Code/Function"),
                             pattern = "\\..R$", full.names = TRUE), source))

simulation_dir <- file.path(base_dir, "Results/Simulated_mortality_rates")
figure_dir <- file.path(base_dir, "Figures/RCP_Simulation/Overall")

age_labels <- c(
  Y20_64 = "20-64",
  Y65_74 = "65-74",
  Y75_84 = "75-84",
  Y_GE85 = "85+"
)

scenario_labels <- c(
  rcp26 = "RCP 2.6",
  rcp85 = "RCP 8.5"
)

scenario_colors <- c(
  "RCP 2.6" = "#007bbd",
  "RCP 8.5" = "#B7282E"
)
```

## Load Simulation Results

```
simulation_results <- list(
  LC = setNames(
```

```

    lapply(names(scenario_labels), function(scenario) {
      load_rcp_simulation("LC", scenario, simulation_dir)
    }),
    scenario_labels
  ),
  LL = setNames(
    lapply(names(scenario_labels), function(scenario) {
      load_rcp_simulation("LL", scenario, simulation_dir)
    }),
    scenario_labels
  )
)

```

## Plot Settings

```

model_configs <- list(
  LC = list(
    output_dir = file.path(figure_dir, "DLNM_LC"),
    filenames = c(
      Athens = "Athens_all_scenarios_no_scale_line.pdf",
      Lisbon = "Lisbon_all_scenarios_no_scale_line.pdf",
      Rome = "Rome_all_scenarios_no_scale_line.pdf"
    ),
    ncol = 2,
    width = 10,
    height = 6,
    x_limits = c(params$start_year, 2046),
    x_breaks = c(2031, 2036, 2041, 2046),
    y_expand = expansion(mult = c(0.2, 0.2)),
    major_y_grid = element_line(color = "gray80"),
    mean_size = 0.4,
    interval_size = 0.3,
    subtitle_size = 14,
    legend_text_size = 11
  ),
  LL = list(
    output_dir = file.path(figure_dir, "DLNM_LL"),
    filenames = c(
      Athens = "Athens_weekly_sim.pdf",
      Lisbon = "Lisbon_weekly_sim.pdf",
      Rome = "Rome_weekly_sim.pdf"
    ),
    ncol = 2,
    width = 10,
    height = 6,
    x_limits = c(params$start_year, 2046),
    x_breaks = c(2031, 2036, 2041, 2046),
    y_expand = expansion(mult = c(0.2, 0.2)),
    major_y_grid = element_line(color = "gray80"),
    mean_size = 0.4,
    interval_size = 0.3,
    subtitle_size = 14,
    legend_text_size = 11
  )
)

```

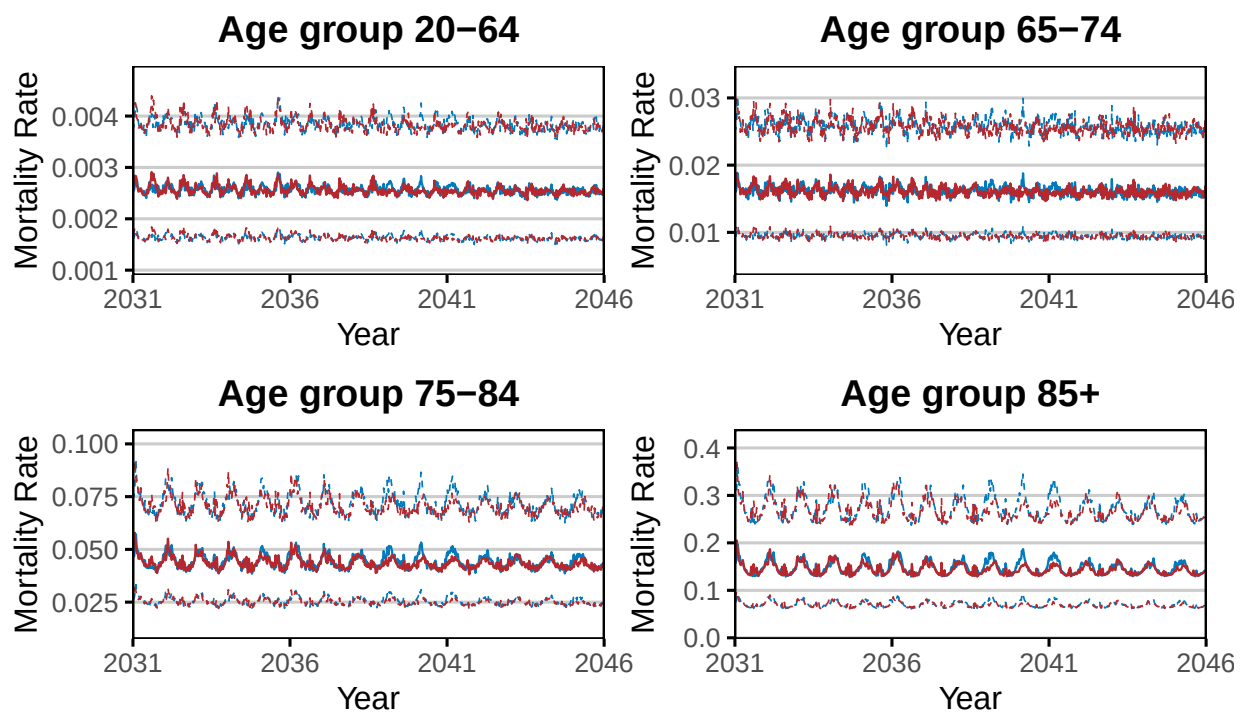
```
)
)
```

## DLNM-LC Visualisation

```
LC_plots <- setNames(lapply(rcp_output_region_names, function(region) {
  plot_rcp_region_model(
    datasets = simulation_results$LC,
    region = region,
    config = model_configs$LC,
    start_index = params$start_index,
    start_year = params$start_year,
    age_labels = age_labels,
    scenario_colors = scenario_colors,
    save_plots = params$save_plots,
    dpi = params$dpi
  )
}), rcp_output_region_names)
```

LC\_plots

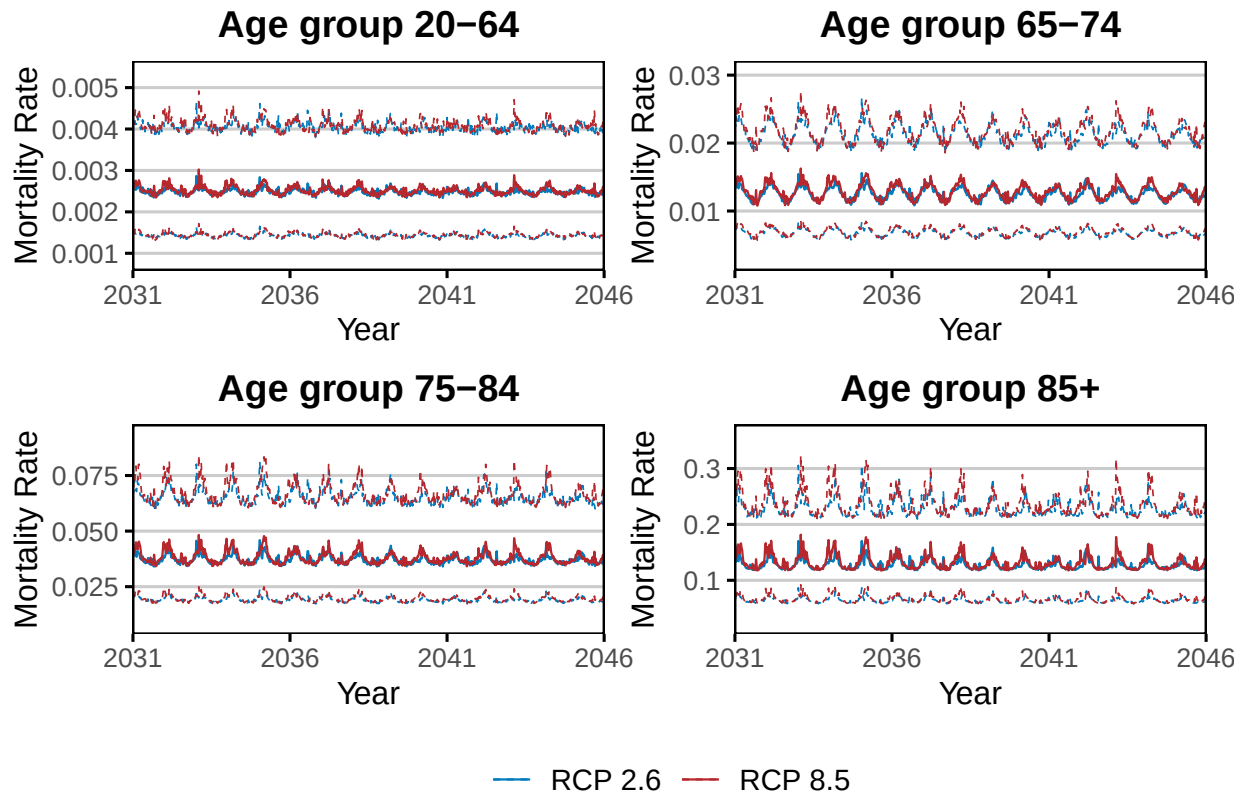
## \$Athens



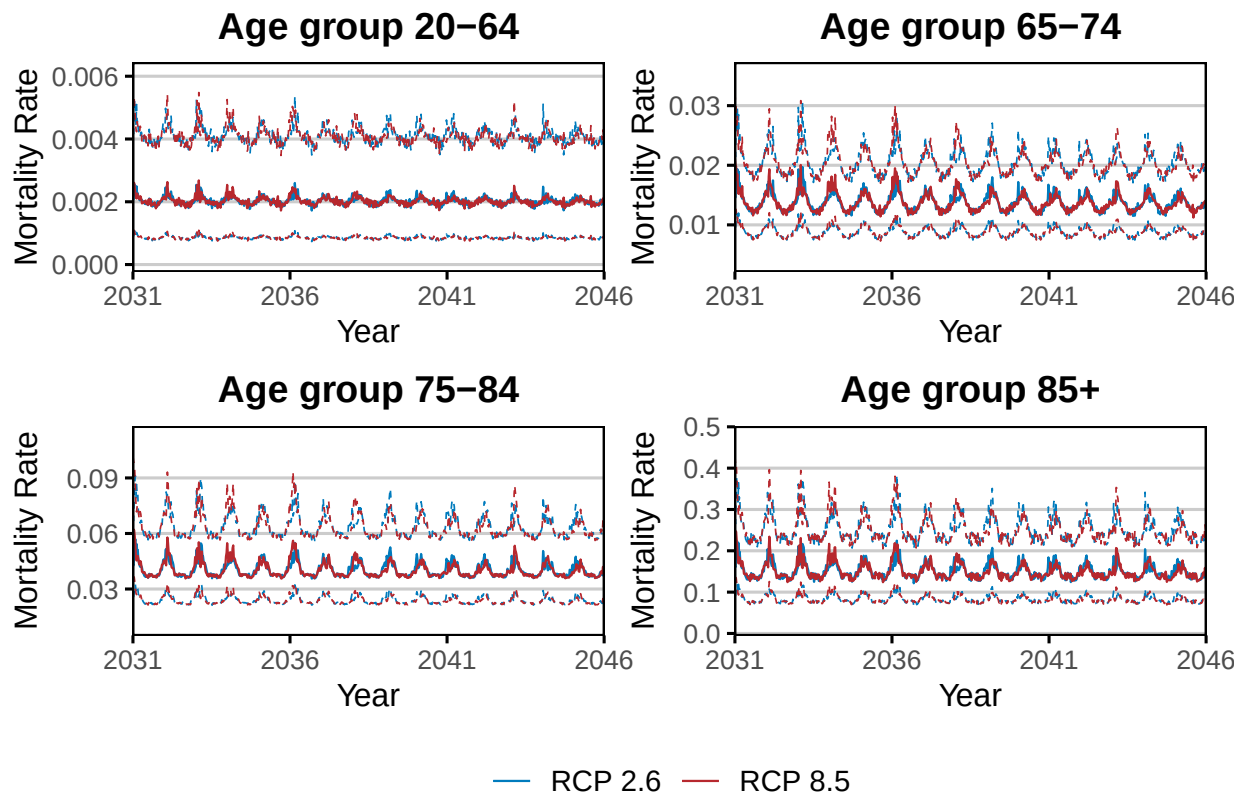
— RCP 2.6 — RCP 8.5

##

## \$Lisbon



##  
## \$Rome

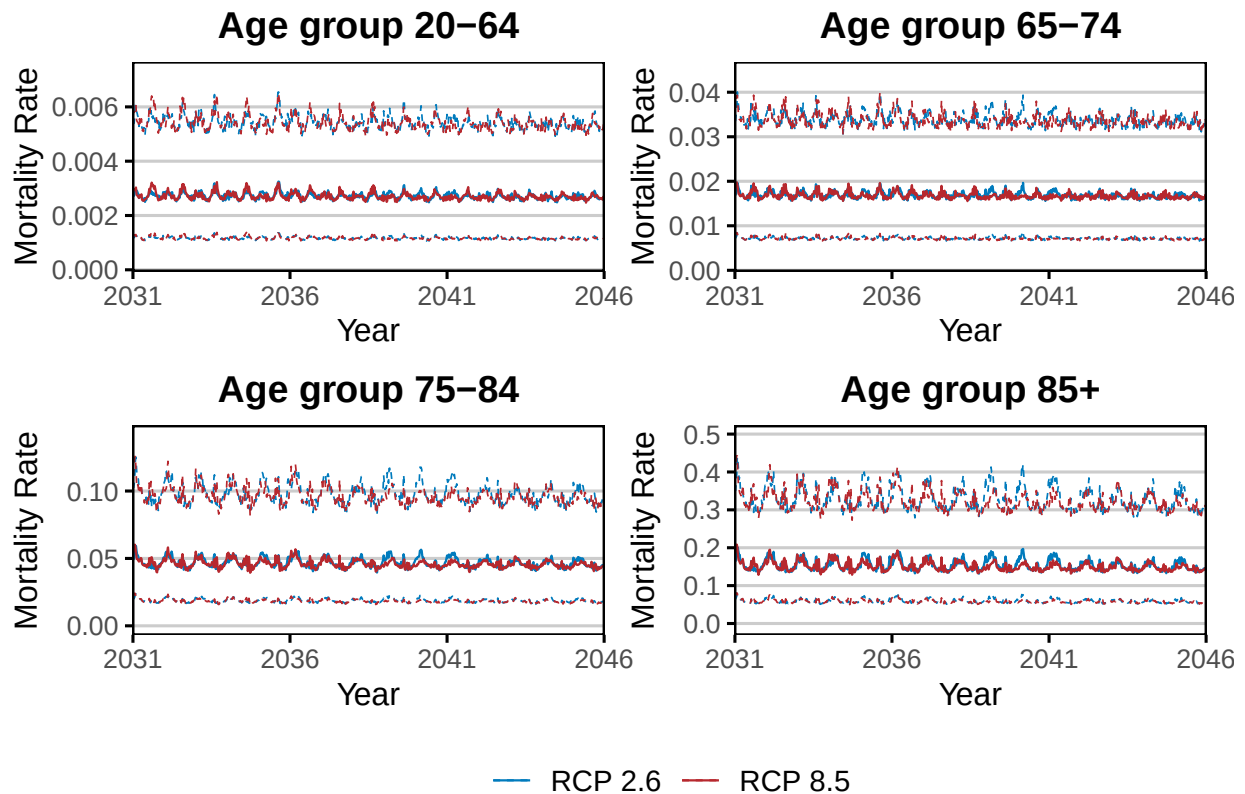


## DLNM-LL Visualisation

```
LL_plots <- setNames(lapply(rcp_output_region_names, function(region) {
  plot_rcp_region_model(
    datasets = simulation_results$LL,
    region = region,
    config = model_configs$LL,
    start_index = params$start_index,
    start_year = params$start_year,
    age_labels = age_labels,
    scenario_colors = scenario_colors,
    save_plots = params$save_plots,
    dpi = params$dpi
  )
}), rcp_output_region_names)

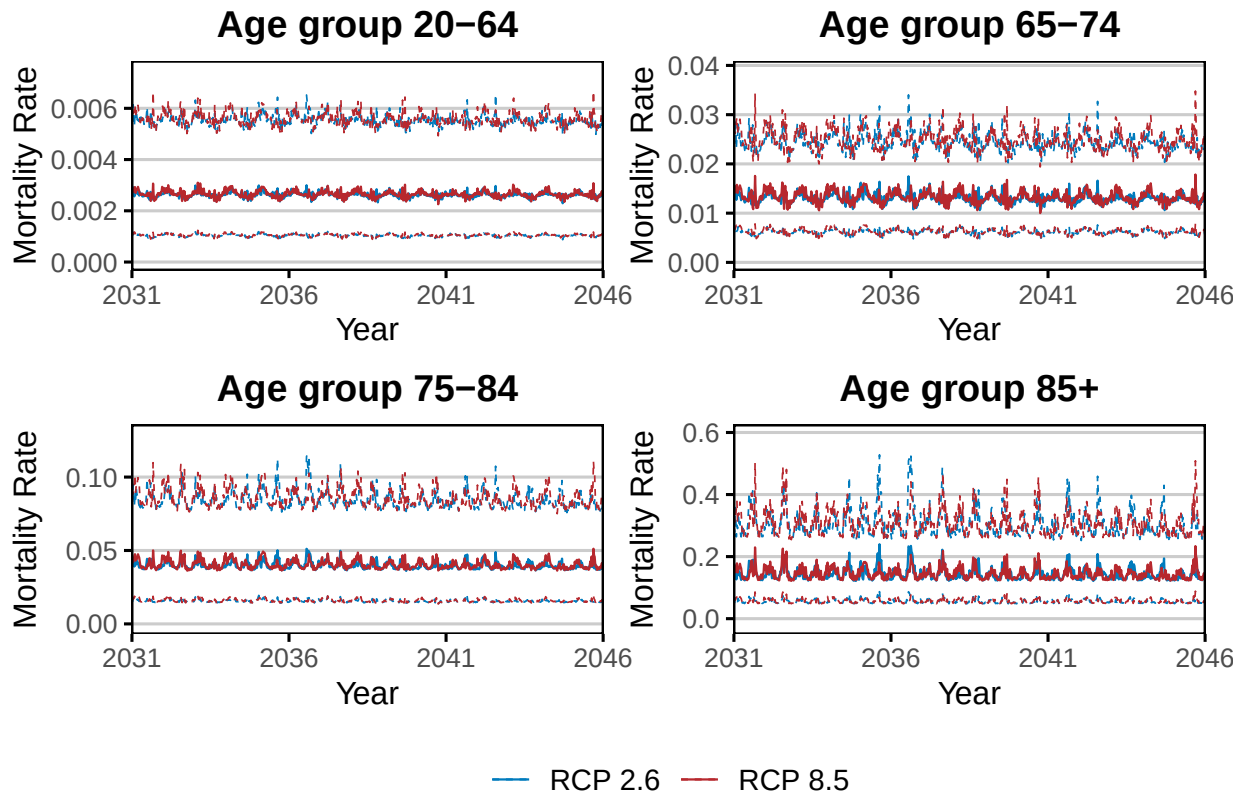
LL_plots
```

## \$Athens



##

## \$Lisbon



##  
## \$Rome

